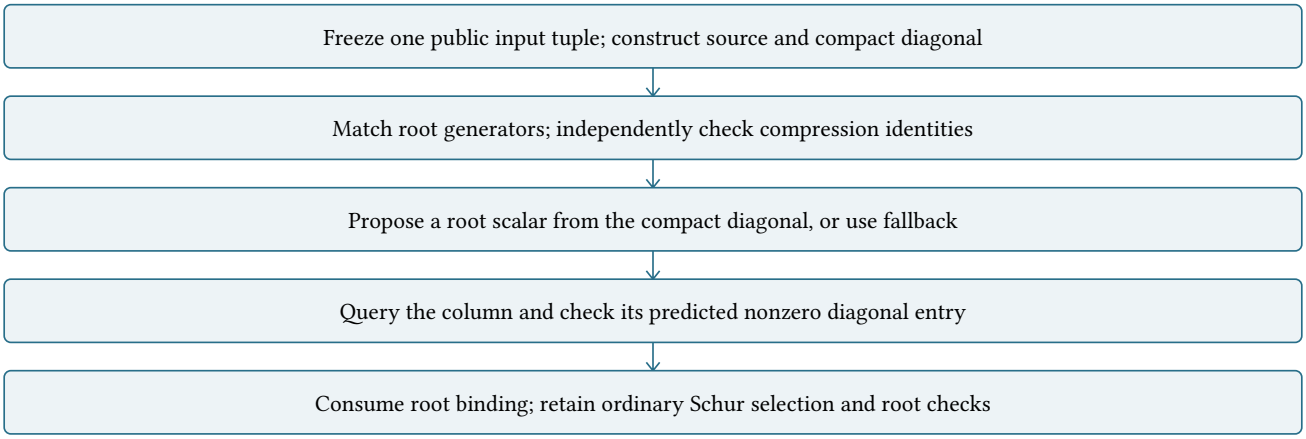
Constructor-bound root diagonal pivots

2026-11-02 · Derived implementation / static checks · Numerical trials: zero

Construction. One immutable input-weight tuple feeds the preserved public source constructor and compact diagonal builder. A binding captures the root source, generators, metric and field signature. Its lifetime ends at the root.



Equality and entry checks. Before rebinding across compression, verify the independent factor certificate. Match the actual new generators to that checked output and require the original metric coordinates. Query the proposed column; its diagonal must match the predicted value and be nonzero. Otherwise fail. This is an implemented runtime gate, not an executed numerical result.

No false prefix. RootSelection is a distinct record with an empty zero prefix and first_column=None. It carries no knowledge of skipped columns. The paired wrapper changes only the admitted record type. If all compact diagonals vanish, invoke the ordinary selector. Never reuse the root packet at a Schur child or after a coordinate permutation.

Derived root proposal	Cost / behavior
Nonzero diagonal found	One full source query + one prediction product
Compact scan	At most q subtractions and q+1 nonzero tests
All diagonals zero	Original selector, with proposal overhead
Root compression check	2ng(s+t) logical products; n(s+2g+t) reductions
Snapshots	2ns then 2nt field entries, plus transient copies

Original compiler, compact-diagonal setup, input copies and r−1 forward-metric products remain charged. Checker integers, Meter field work and native comparisons are separate units. No runtime or total-heap improvement is measured.

Changed pivots. A later scalar may replace the original selector's pair, changing the downstream pivot tree. Validate final root equations and normalized witness pairing, not byte equality of nonunique vectors or identical pivots. Early-witness rank semantics and separate feature-annihilation checks remain.

Verification / next. Two complete static AST comparisons cover driver and paired transport integration; six Python files parse. Defaults and gate ordering are statically checked. Numerical/corruption/heap/performance trials remain zero under the existing cost gate. Funded controls include altered generators, certificates, weights, fields, views and predictions; fallback and zero source; both outcomes, counters and lifetime. Next derive charged Schur diagonal updates from accepted pivot columns, with explicit metric and storage accounting.

Sources. DERIVATION.md, root_binding.py, driver_root.py, root_move.py, public_solve.py and integration.json. Established scalar/pair pivot context: Dumas–Pernet2018, arXiv:1802.10453, introduction and Section 2.